

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



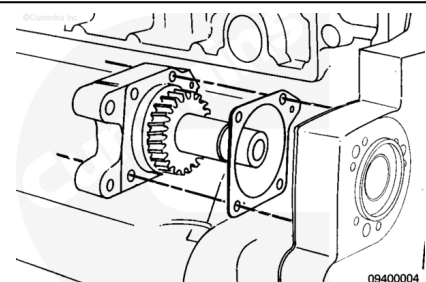
- Clean the water pump with solvent.
- Remove the alternator. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/20/20-013-001-tr.html>).
- Remove the belt guard. Refer to Procedure 008-001 (</qs3/pubsys2/xml/en/procedures/20/20-008-001.html>).
- Remove the alternator belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/20/20-013-005-tr.html>).
- Remove the water pump drive pulley. Refer to Procedure 009-032 (</qs3/pubsys2/xml/en/procedures/20/20-009-032.html>).
- Remove the alternator drive oil seal. Refer to Procedure 001-001 (</qs3/pubsys2/xml/en/procedures/20/20-001-001-tr.html>).

Remove

Remove the water pump drive assembly.

Remove and discard the gasket.

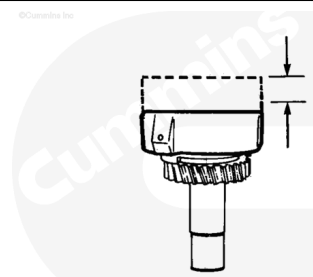
Note : If necessary, tap on the end of the shaft with a mallet.



Inspect for Reuse

Measure the water pump drive end clearance.

mm		in
0.23	MIN	0.009
0.33	MAX	0.013

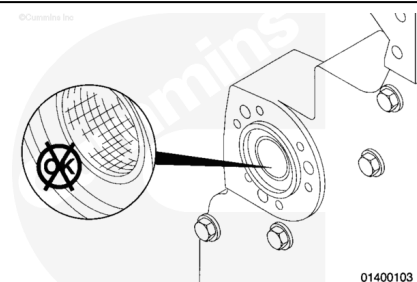


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If the end clearance is **not** within specifications, replace or rebuild the assembly.

Note : Oversize thrust bearings are available to adjust the end clearance.

Inspect the water pump drive bushing in the front cover for excessive wear or damage. If replacement is required refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html>).



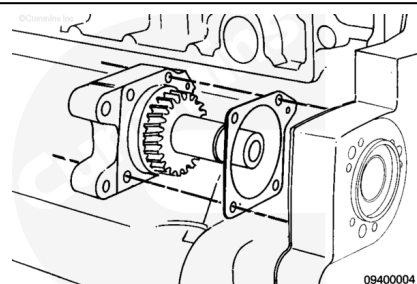
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Install

Install the gasket. Install the water pump drive assembly.

The drive will be secured in position by the stud.

Water Pump Drive Capscrew	45 n.m	[33 ft-lb]
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Finishing Steps

- Install the water pump. Refer to Procedure 008-062 (/qs3/pubsys2/xml/en/procedures/20/20-008-062-tr.html).
- Install the alternator drive oil seal. Refer to Procedure 001-001 (/qs3/pubsys2/xml/en/procedures/20/20-001-001-tr.html).
- Install the water pump drive pulley. Refer to Procedure 009-032 (/qs3/pubsys2/xml/en/procedures/20/20-009-032.html).
- Install the alternator belt. Refer to Procedure 013-005 (/qs3/pubsys2/xml/en/procedures/20/20-013-005-tr.html).
- Install the belt guard. Refer to Procedure 008-001 (/qs3/pubsys2/xml/en/procedures/20/20-008-001.html).
- Operate the engine and check for leaks.

